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ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-2** **Date: 30-03-2026**

Time: 03.00Pm to 06.00Pm **GTM-33** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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41
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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let $A=\left\{\theta\in[0,2\pi]:1+10\,\mathrm{Re}\left(\frac{2\cos\theta+i\sin\theta}{\cos\theta-3i\sin\theta}\right)=0\right\}$,

Then $\sum_{\theta\in A}4\theta^2$ is equal to

1) $8\pi^2$ **2)** $21\pi^2$ **3)** $\frac{27\pi^2}{4}$ **4)** $6\pi^2$

2. If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440th position in this arrangement, is :

1) PRNAKU **2)** PRKANU **3)** PRKAUN **4)** PRNAUK

3. The sum of the coefficients of $x^{\frac{2}{3}}$ and $x^{\frac{-2}{5}}$ in the binomial expansion of $\left(x^{\frac{2}{3}}+\frac{1}{2}.x^{\frac{-2}{5}}\right)^9$ is

1) $\frac{21}{4}$ **2)** $\frac{69}{16}$ **3)** $\frac{63}{16}$ **4)** $\frac{19}{4}$

4. The number of terms of an A.P is even; the sum of all the odd terms is 24, the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is

1) 8 **2)** 10 **3)** 6 **4)** 4

5. Consider the following two statements:

Statement 1: For any two nonzero complex numbers Z_1, Z_2

$\left(|Z_1|+|Z_2|\right)\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right|\le 2\left(|Z_1|+|Z_2|\right)$ and

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Statement 2: If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$, $then\ \frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y}=1$.

Between the above two Statements;

1) Both statement 1 and statement 2 are incorrect

2) Statement 1 is incorrect but statement 2 is correct

3) Statement 1 is correct but statement 2 is incorrect

4) Both statement 1 and statement 2 are correct

6. If $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+...+\frac{1}{\sqrt{99}+\sqrt{100}}=m$ and $\frac{1}{1.2}+\frac{1}{2.3}+...+\frac{1}{99.100}=n$ then the point (m, n) lies on the line

1) $11(x-1)-100(y-2)=0$ **2)** $11(x-2)-100(y-1)=0$

3) $11(x-1)-100y=0$ **4)** $11x-100y=0$

7. Let $y^2=12x$ the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ)=\frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2+64y^2-\alpha x-64\sqrt{3}y=\beta$, then $\frac{\beta-\alpha}{664}$ is equal to

1) 1 **2)** 2 **3)** 3 **4)** 4

8. If $L=\lim\limits_{x\to 0^+}\left(\frac{\tan\left(5(x)^{\frac{1}{3}}.\log_e\left(1+3x^2\right)\right)}{\left(\tan^{-1}\left(3\sqrt{x}\right)\right)^2.\left(e^{5(x)^{\frac{4}{3}}}-1\right)}\right)$ then 24L is equal to

1) 3 **2)** $\frac{1}{3}$ **3)** 8 **4)** $\frac{1}{8}$

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9. The mean and standard deviation of 100 observations are 40 and 5.1 respectively, by mistake one observation is taken as 50 instead of 40. If the correct mean and the correct standard deviation are μ and σ respectively, then $20(\mu+\sigma)$ is equal to

1) 445 **2)** 451 **3)** 898 **4)** 449

10. Match the statements /expressions given in List I with the values given in List II.

List-I	List-II
a) In R^2, if the magnitude of the projection vector of the vector \ $\alpha i+\beta j$ on $\sqrt{3}i+j$ is $\sqrt{3}$ and if $\alpha=2+\sqrt{3}\beta$, then the possible value(s) of $\lvert\alpha\rvert$ is (are)	p) 1
b) Let a and b be real numbers such that the function $f(x)=\begin{cases}-3ax^2-2, & x<1\\ bx+a^2, & x\geq 1\end{cases}$ is differentiable for all $x\in R$. Then the possible value(s) of $\lvert a\rvert$ is (are)	q)2
c) Let $w\neq 1$be a complex cube root of unity. If $\left(3-3w+2w^2\right)^{4x+3}+\left(2+3w-3w^2\right)^{4x+3}+\left(-3+2w+3w^2\right)^{4x+3}=0$, then possible values(s) of x is (are)	r)3
d) Let the harmonic mean of two positive real numbers 'a' and 'b' be 4. If q is a positive real number such that a, 5, q, b is an arithmetic progression , then value(s) of $\lvert q-a\rvert$ is (are)	s) 4
	t) 5

1) $a\to p,r;b\to p,t;c\to p,q;d\to s,t$

2) $a\to p,q;b\to p,q;c\to p,q,s,t;d\to q,t$

3) $a\to q,s;b\to p,r;c\to p,t;d\to p,t$

4) $a\to p,t;b\to p,s;c\to s,t;d\to r,t$

Sri Chaitanya Educational Institutions Infinity Learn 41
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11. If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is:

1) $\frac{965}{2^{11}}$ **2)** $\frac{965}{2^{10}}$ **3)** $\frac{945}{2^{11}}$ **4)** $\frac{945}{2^{10}}$

12. Let $f(x)=\log_e^x$ and $g(x)=\frac{x^4-2x^3+3x^2-2x+2}{2x^2-2x+1}$ Then the domain of fog is

1) R **2)** $(0,\infty)$ **3)** $[0,\infty)$ **4)** $[1,\infty)$

13. Let $S=\left\{x:\cos^{-1}x=\pi+\sin^{-1}x+\sin^{-1}(2x+1)\right\}$.

Then $\sum_{x\in s}(2x-1)^2$ is equal to_____

1) 3 **2)** 5 **3)** 15 **4)** 25

14. Let $B=\begin{bmatrix}1 & 3\\ 1 & 5\end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1}=A^{-1}$. If $BCB^{-1}=A$ and

$c^4+\alpha c^2+\beta I=0, then (2\beta-\alpha)^2$ is equal to

1) 10 **2)** 5 **3)** 100 **4)** 64

15. Let $A=\begin{bmatrix}2 & 2+p & 2+p+q\\ 4 & 6+2p & 8+3p+2q\\ 6 & 12+3p & 20+6p+3q\end{bmatrix}$. If $\det\left(adj\left(adj\left(3A\right)\right)\right)=2^m.3^n; m,n\in N,$

Then $m\times n$ is equal to

1) 120 **2)** 12 **3)** 24 **4)** 144

16. Let $f(x)$ be a real differentiable function such that $f(0)=1$ and

$f(x+y)=f(x).f'(y)+f'(x).f(y)$ for all $x,y\in R$. Then $\sum_{n-1}^{100}\log_e f(n)$ is equal to:

1) 2406 **2)** 2525 **3)** 2384 **4)** 5220

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17. If $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2,\ \forall x\neq 0$ and $y=9x^2f(x)$ then y is strictly increasing in:

1) $\left(0,\frac{1}{\sqrt{5}}\right)U\left(\frac{1}{\sqrt{5}},\infty\right)$ **2)** $\left(\frac{-1}{\sqrt{5}},0\right)U\left(\frac{1}{\sqrt{5}},\infty\right)$

3) $\left(\frac{-1}{\sqrt{5}},0\right)U\left(0,\frac{1}{\sqrt{5}}\right)$ **4)** $\left(-\infty,\frac{1}{\sqrt{5}}\right)U\left(0,\frac{1}{\sqrt{5}}\right)$

18. If $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3.x^{-24}+x^{-26}}\right)dx=\frac{\alpha}{3(\alpha+1)}\left(3.x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+c;$

$x>0,(\alpha,\beta,\gamma\in z)$, where C is the constant of integration , then $\alpha^2+\beta^2+\gamma^2$ is equal to

1) 23 **2)** 19 **3)** 539 **4)** 529

19. If $10\sin^4\theta+15\cos^4\theta=6$, then the value of $625\left(\frac{-27\,\mathrm{cosec}^6\theta+8\sec^6\theta}{16\sec^8\theta}\right)$ *is*;

1) 50 **2)** 0 **3)** $\frac{5}{2}$ **4)** $\frac{2}{5}$

20. If the range of the function $f(x)=\frac{5-x}{x^2-3x+2},x\neq 1,2$ is $(-\infty,\alpha]U[\beta,\alpha)$ then $\frac{\alpha^2+\beta^2}{194}$, equal to

1) 63 **2)** 47 **3)** 194 **4)** 1

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

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21. The value of $\dfrac{e^{\frac{-\pi}{4}}+\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{50}x\right)dx}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x+\tan^{51}x\right)dx}$ is _____

22. The area of the region $\left\{(x,y):|x-y|\le y\le 4\sqrt{x}\right\}$ is A then $\dfrac{3A}{512}$ is _____

23. Let $y=y(x)$ be the solution of the D.E. $\left(x^2+1\right)y'-2xy=\left(x^4+2x^2+1\right)\cos x, y(0)=1$

Then $\int_{-3}^{3} y(x)dx$ is _____

24. Let the position vectors of three vertices of a triangle be $4\bar{p}+\bar{q}-3\bar{r}$, $-5\bar{p}+\bar{q}+2\bar{r}$, and $2\bar{p}-\bar{q}+2\bar{r}$. If the position vectors of the orthocentre and the Circumcentre of the triangle are $\dfrac{\bar{p}+\bar{q}+\bar{r}}{4}$ and $\alpha\bar{p}+\beta\bar{q}+\gamma\bar{r}$ respectively, then $\dfrac{1}{3}(\alpha+2\beta+5\gamma)$ is equal to _____

25. Let the shortest distance between the lines

$\dfrac{x-3}{3}=\dfrac{y-\alpha}{-1}=\dfrac{z-3}{1}$ and $\dfrac{x+3}{-3}=\dfrac{y+7}{2}=\dfrac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive of $\dfrac{3(5\alpha+\beta)}{46}$ is _____

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Assertion: The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason: Nuclear force is short-ranged.

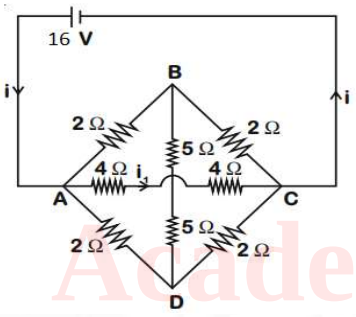
1) If both Assertion and Reason are true and the reason is correct explanation of the Assertion.

2) If both Assertion and Reason are true and the reason is not correct explanation of the Assertion.

3) If Assertion is true and Reason is false

4) If Assertion is false and Reason is true

27. The value of current i_1 flowing from A to C in the circuit diagram is:



1) 5 A **2)** 2.25 A **3)** 0.75 A **4)** 2 A

28. Assertion: A bar magnet dropped through a metallic cylindrical pipe takes more time to come down compared to a non-magnetic bar with the same geometry and mass.

Reason: For the magnetic bar, Eddy currents are produced in the metallic pipe which opposes the motion of the magnetic bar.

In the light of the above statements, choose the correct the correct answer from the options given below

1) If both assertion and Reason are true and the reason is correct explanation of the Assertion.

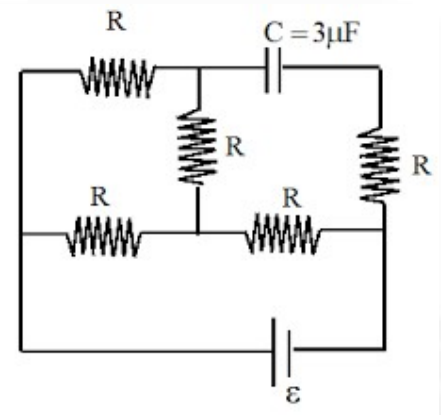
2) If both Assertion and Reason are true and the reason is not correct explanation of the Assertion.

3) If assertion is true and Reason is false

4) If Assertion is false and Reason is true

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JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
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29. In the given circuit, the potential difference across the capacitor is 12 V. Each resistance is of 3Ω. The cell is ideal. The emf of the cell is



1) 15 V **2)** 9 V **3)** 12 V **4)** 24 V

30. Statement-I: A balloon filled with hydrogen will fall an acceleration $\frac{g}{6}$ on the moon

Statement-II: Moon has no atmosphere.

1) Statement-I is false but Statement-II is true

2) Statement-I is true but Statement-II are false

3) Both Statement-I and Statement-II are true

4) Both Statement-I and Statement-II are false

31. Statement-I: When a prism of $\mu = 3/2$ is immersed in water $(\mu = 4/3)$, deviation through the prism becomes $1/4^{th}$ of the deviation, when the prism is in air.

Statement-II: It follows from $\delta = (\mu - 1)A$

1) Statement-I is true, statement-II is true. Statement-II is correct explanation of statement-I

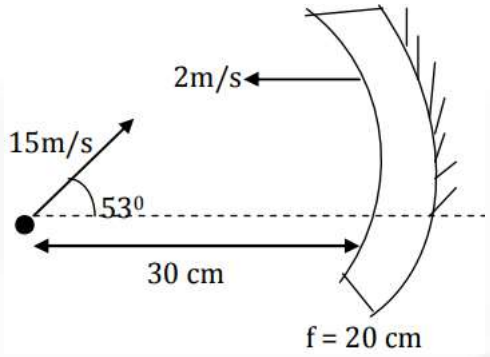
2) Statement-I is true, statement-II is true, but statement-II is not correct explanation of statement-I

3) Statement-I is true, statement-II is false

4) Statement-I false, statement-II is true

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32. Find the velocity of image in a situation as shown in the figure.



1) $\left(-46\hat{i}-20\hat{j}\right)\frac{m}{s}$ **2)** $\left(-24\hat{i}-46\hat{j}\right)\frac{m}{s}$ **3)** $\left(-46\hat{i}-24\hat{j}\right)\frac{m}{s}$ **4)** $\left(-40\hat{i}-24\hat{j}\right)\frac{m}{s}$

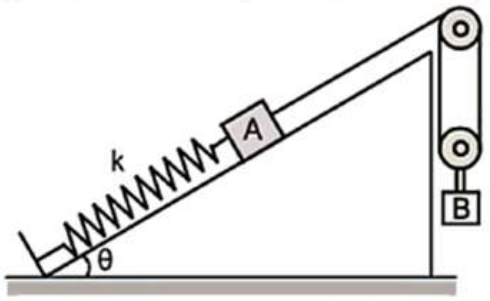
33. A closed coil having 50 turns, area $300cm^2$, is rotated from a position where it plane makes an angle of 45^0 with a magnetic field of flux density 2.0T to a position perpendicular to the field in a time of 0.1s. What is the average EMF induced in the coil?

1) 11.8 V **2)** 19.8 V **3)** 8.8 V **4)** 4.8 V

34. A solid copper sphere cools at the rate of 2.8^0C per minute, when its temperature is 127^0C. Find the rate at which another solid copper sphere of twice the radius lose its temperature at 327^0C, if in both the cases, the room temperature is maintained at 27^0C.

1) 4.5^0C / min **2)** 9.72^0C / min **3)** 1.2^0C / min **4)** 3.2^0C / min

35. Calculate the angular frequency of the system shown in figure. Friction is absent everywhere and the threads, spring and pulleys are massless. Given that $m_A = m_B = m$



1) $\omega = \sqrt{\frac{3k}{m}}$ **2)** $\omega = \sqrt{\frac{3k}{7m}}$ **3)** $\omega = \sqrt{\frac{4k}{5m}}$ **4)** $\omega = \sqrt{\frac{7k}{5m}}$

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JEE Advanced 2023 RANK 1
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36. The pressure of one mole of an ideal gas varies according to the law $p = p_o - \alpha V^2$ where p_o and α are positive constants. The highest temperature that, the gas may attain is

1) $\frac{2p_0}{3R}\left(\frac{p_0}{3\alpha}\right)^{1/2}$ **2)** $\frac{3p_0}{2R}\left(\frac{p_0}{3\alpha}\right)^{1/2}$ **3)** $\frac{p_0}{R}\left(\frac{p_0}{3\alpha}\right)^{1/2}$ **4)** $\frac{p_0}{R}\left(\frac{p_0}{\alpha}\right)^{1/2}$

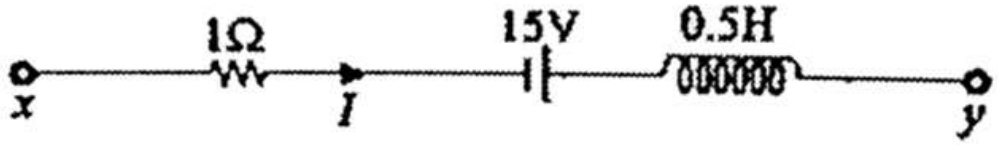
37. Match the value with its number of significant figures

Column I		Column II	
(1)	2.0040×10^4	(p)	4
(2)	0.0052125	(q)	5
(3)	3.010×10^3	(r)	3
(4)	0.00628	(s)	2
		(t)	7

1) $A\to q;B\to q;C\to p;D\to r$ **2)** $A\to t;B\to q;C\to r;D\to p$

3) $A\to q;B\to q;C\to s;D\to r$ **4)** $A\to t;B\to q;C\to p;D\to s$

38. A section of an electrical circuit XY is shown in figure which carries a current of 5A which is decreasing at the rate of $10A/s$. Find the potential difference $V_y - V_x$



1) $V_y - V_x = 10V$ **2)** $V_y - V_x = 20V$ **3)** $V_y - V_x = 5V$ **4)** $V_y - V_x = 15V$

39. A double slit experiment is immersed in a liquid of refractive index 1.33 It has slit separation of 1 mm and distance between the plane of slits and screen 1.33 mm. The slits are illuminated by a parallel beam of light whose wavelength in air is $6830A^0$ calculate the fringe width.

1) $3.3\times10^{-5}m$ **2)** $4\times10^{-2}m$ **3)** $6.3\times10^{-4}m$ **4)** $8.3\times10^{-9}m$

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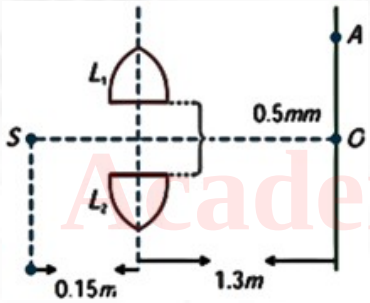
40. A parallel plate capacitor consists of two circular plates each of radius 2cm, separated by a distance of 0.1 m if voltage across the plates is varying at the rate of $5\times10^{13}Vs^{-1}$, then the value of displacement current is

1) $5.56\,A$ **2)** $5.56\times10^{-2}\,A$ **3)** $5.56\times10^{3}\,A$ **4)** $2.28\times10^{4}\,A$

41. Two identical non relativistic particles move at right angles to each other, possessing de-Broglie wavelengths λ_1 and λ_2. Find the de-Broglie wavelength of each particle in the frame of their centre of mass.

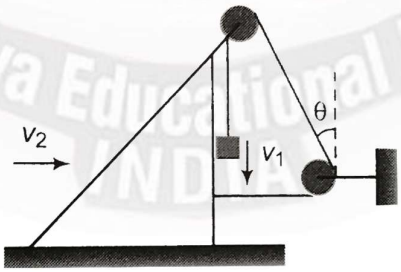
1) $\lambda_1+\lambda_2$ **2)** $\dfrac{\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$ **3)** $\dfrac{\lambda_1\lambda_2}{\sqrt{\lambda_1+\lambda_2}}$ **4)** $\dfrac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$

42. In the given figure, S is monochromatic point source emitting light of wave length $\lambda=500nm$. A thin lens of circular shape and focal length 0.1m is cut into two identical halves L_1 and L_2 by a plane passing through diameter. The two halves are symmetrically placed about the central axis SO with gap of 0.5mm. The distance along the axis from S to L_1 and L_2 is 0.15m, while that from L_1 and L_2 to 'O' is 1.3m. The screen at O is normal to SO. If the third maxima occurs at point A on the screen find the distance OA?



1) 1mm **2)** 2mm **3)** 1.5mm **4)** 2.5mm

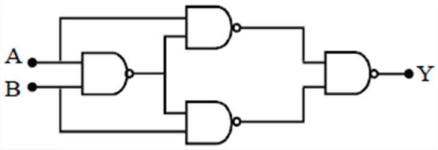
43. In the figure block moves downwards with velocity v_1, the wedge moves rightwards with velocity v_2. The correct relation between v_1 and v_2 is



1) $v_2=v_1$ **2)** $v_2=v_1\sin\theta$ **3)** $2v_2\sin\theta=v_1$ **4)** $v_2(1+\sin\theta)=v_1$

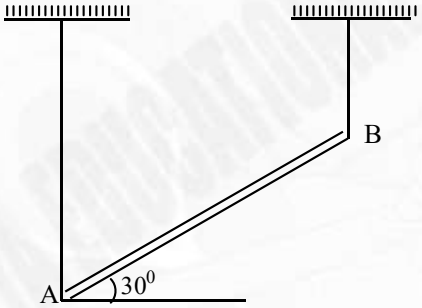
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44. Select the outputs Y of the combination of gates shown below for inputs $A=1,B=0;A=1,B=1$ and $A=0,B=0$ respectively



1) (011) **2)** (001) **3)** (100) **4)** (111)

45. A thin uniform bar AB of mass M and length 2L is held at an angle of 30^0 to the horizontal by means of two vertical strings at each end. The string at the right end breaks. Find tension in the left string immediately after the right string breaks.



1) $t=\frac{Mg}{7}$ **2)** $T=\frac{Mg}{3}$ **3)** $T=\frac{3Mg}{5}$ **4)** $T=\frac{4Mg}{13}$

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

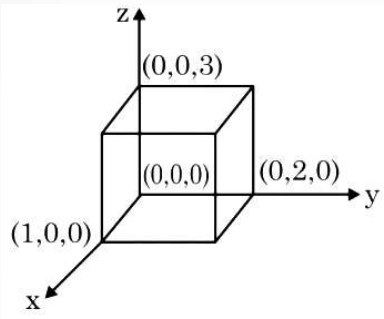
46. Suppose a uniformly charged wall provides a uniform electric field of $2\times10^7\,NC^{-1}$ normally. A charged particle of mass 2 g being suspended through a silk thread of length 20cm and remain stayed at a distance of 10cm from the well. Then the charge on the particle will be $\frac{1}{\sqrt{x}}nC$ where $x=$_____ [Use $g=10ms^{-2}$]

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47. As shown in figure, a cuboid lies in a region with electric field $E = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}\ N/C$.

The magnitude of charge with in the cuboid is $n \in_0 c$. The value of n is_____

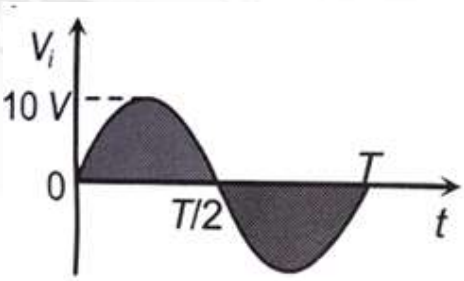
(If of cuboid has dimensions $1 \times 2 \times 3m^3$)

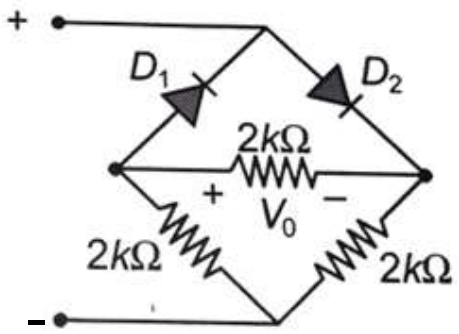


48. A small disc and a thin uniform rod of length L, whose mass is η times greater than the mass of the disc, lie on a smooth horizontal plane. The disc is set in motion, in horizontal direction and perpendicular to the rod, with velocity v, after which it elastically collides with the end of the rod. Find the velocity of the disc and the angular velocity of the rod after the collision. At what value of η will the velocity of the disc after the collision be equal to zero?

49. A wire of length 1.0 m and radius 10^{-3} m is carrying a heavy current and is assumed to radiate as a black body. At equilibrium temperature of wire is 900K while that of the surroundings is 300K. The resistivity of the material of the wire at 300K is $\pi^2 \times 10^{-8}\Omega - m$ and its temperature coefficient of resistance is $7.8 \times 10^{-3}/C$. Find the current in the wire is_____ A $\left[\sigma \cong 5.68 \times 10^{-8} wm^{-2}K^{-4}\right]$

50. In the circuit shown in figure the maximum output voltage V_0 is _____V





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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Which of the following are the correct Statements?

A) To avoid the medical condition "Bends ", the tanks used by scuba divers are filled with air diluted with helium.

B) In case of negative deviation from Rault's law, the intermolecular attractions between A-A and B-B are weaker than those between A-B

C) The relative lowering of vapour pressure in a solution containing a non-volatile solute is equal to the molefraction of the solvent

D) When the isotonic solutions are separated by a semipermeable membrane, no osmosis occurs between them

E) The van't Hoff factor (i) for 80% dissociation of the complex $K_4\left[Fe(CN)_6\right]$ is 4.2

1) A,B,C,D,E **2)** A,B,D,E **3)** A,C,D **4)** B,C,D

52. $NaCl + AgNO_3 \rightarrow \underset{(\text{white})}{A\downarrow} + B$

$A + NH_{3(solution)} \rightarrow \underset{(\text{soluble complex})}{C}$

$CuSO_4 + NH_{3(solution)} \rightarrow \underset{(\text{soluble complex})}{D}$

Select the correct statements

A) IUPAC name of complex D is Tetraamminecopper (II) sulphate

B) Central atom in complex C is 'sp' hybridised

C) By adding HNO_3, complex C further converts into AgCl

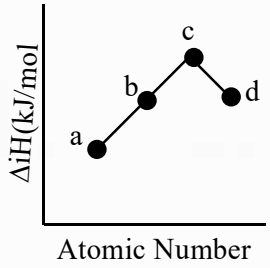
D) Spin only magnetic moment in complex D is Zero

1) Only D **2)** only A and B

3) only A, B and C **4)** A, B, C and D

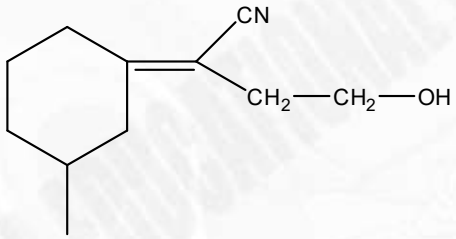
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53. If a, b, c and d are consecutive elements in the second period of modern periodic table, identity b and d. (ΔiH is first ionisation enthalpy)



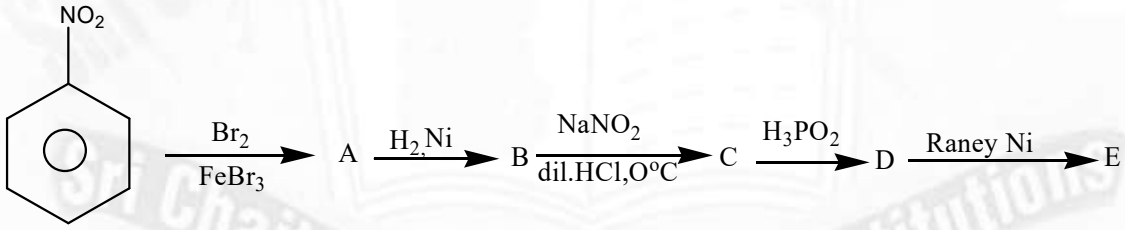
1) C, O **2)** Be, C **3)** B, N **4)** N, F

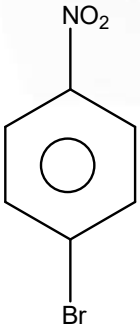
54. False statement about the following compound is

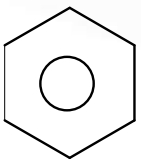


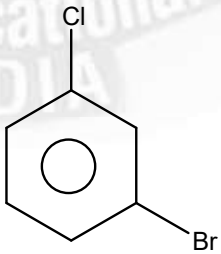
1) It is having two stereo genic carbons
2) It shows optical isomerism
3) Its configuration at double bond is E
4) It shows geometrical isomerism

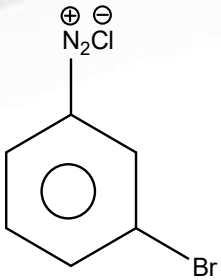
55. The end product E is



1) 

2) 

3) 

4) 

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56. Statement-1: C_2 gas molecule consists of a double bond with two Pi bonds (as per MOT)

Statement-2: Dipole moment of NH_3 is more than that of NF_3

1) Both Statement -1 and Statement-2 are Correct

2) Both Statement -1 and Statement -2 are incorrect

3) Statement -1 is correct but Statement -2 is incorrect

4) Statement -1 is incorrect but statement -2 is correct

57. The combustion of one mole of benzene takes place at 298K and 1 atm. After combustion, $CO_2(g)$ and $H_2O(\ell)$ are produced and 3267.0 kJ of heat is liberated. Standard enthalpies of formation of $CO_2(g)$ and $H_2O(\ell)$ are -393.5 kJ mol^{-1} and -285.83 kJ mol^{-1} respectively. Then standard enthalpy of formation of benzene is -------- kJ mol^{-1}

1) -48.51 **2)** 3267 **3)** -3267 **4)** 48.51

58. Match the following sets 1 and 2

Set-1	Set-2
A) The maximum number of spectral lines possible when the electron in hydrogen atom jumps from 4 to 1 orbits	1) 9
B) The number of radial nodes possible for 4s orbital	2) 3
C) The number of degenerate orbitals in the M-shell of hydrogen atom	3) 3.5
D) The sum of four quantum numbers(n, l, m, s) for the valence electron in Na atom	4) 6

1) A-4, B-2, C-3, D-1 **2)** A-4, B-2, C-3, D-3

3) A-4, B-2, C-1, D-3 **4)** A-4, B-2, C-1, D-1

59. In a reaction, $A + B_2 \rightarrow AB_2$, identity the limiting reagent and find the number of molecules of AB_2 formed when 300 atoms A combine with 200 molecules of B.

1) A,300 **2)** B_2,300 **3)** B_2,200 **4)** A,200

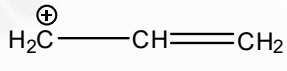
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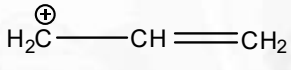
60. Match the following Lists I & II

List-I	List-II
A) Atomic radius	1) $B > Al > Tl > In > Ga$
B) Ionisation enthalpy(ΔiH_1)	2) $Al < Ga < In < Tl < B$
C) Electronegativity	3) $B > Tl > Ga > Al > In$
D) Melting point	4) $B < Ga < Al < In < Tl$

1) A-1, B-2, C-3, D-4 **2)** A-1, B-4, C-3, D-2

3) A-4, B-2, C-3, D-1 **4)** A-4, B-3, C-2, D-1

61. Assertion (**A**) :  is more stable carbocation then 

Reason (R) : +M effect in  gives octet configuration to both C and N but it is absent in 

1) Both A and R are incorrect

2) A is correct but R is incorrect

3) Both A and R are correct and R is the correct explanation to A

4) Both A and R are correct but R is not the correct explanation to A

62. Assertion (**A**) : Fe^{3+} catalyses the reaction between iodide and perdisulphate ions.

Reason (R) : Fe^{3+} oxidises I^- into I_2 first and then Fe^{2+} reduces $S_2O_8^{2-}$ into SO_4^{2-}

1) Both A and R are correct and R is the correct explanation to A

2) Both A and R are correct and R is not the correct explanation to A

3) A is correct but R is incorrect

4) A is incorrect but R is correct

63. To an aqueous solution containing $Al^{3+}, Zn^{2+}, Ca^{2+}, Fe^{3+}, Ni^{2+}, Ba^{2+}, Ag^+$ and Cu^{2+}, $conc.HCl$ was added followed by H_2S. The total number of cations precipitated during this reaction is/are

1) 1 **2)** 3 **3)** 2 **4)** 4

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64. Identity the incorrect statement among the following.

1) Amylopectin is insoluble in water and constitutes about 80 – 85% of starch

2) During denaturation of a protein, its primary structure is destroyed but secondary and tertiary structures remain intact.

3) Vitamin-E deficiency causes increased fragility of RBC_s and muscular weakness.

4) Nucleic acid (RN**A)** contains B-D-ribose, Adenine, Guanine Cytosine and uracil as the Structural components.

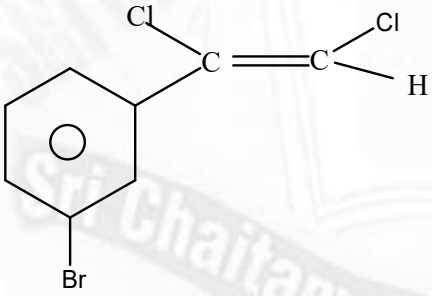
65. In a first order decomposition reaction, the time taken for the decomposition of reactant to one half and one fourth of its initial concentration are t_1 and t_2 respectively, the ratio of $\frac{t_1}{t_2}$ will be:

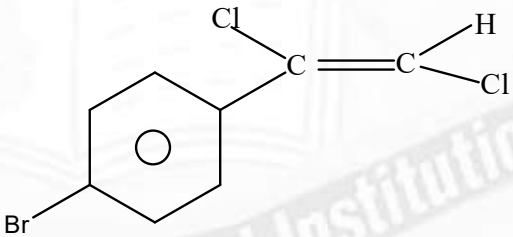
1) $\frac{1}{2}$ **2)** $\frac{1}{4}$ **3)** 4 **4)** 2

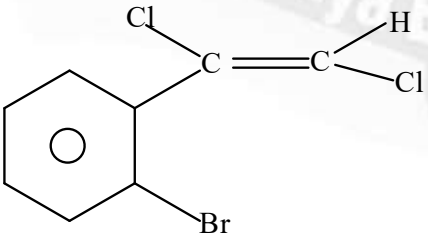
66. Product A in the following sequence of reactions

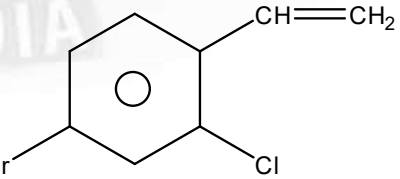
Ethyl benzene $\xrightarrow[(b)Cl_2,\Delta \\ (c)alc.KOH \\ (d)Cl_2,CCl_4]{(a)Br_2,Fe} A(major\ product)$

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1) 

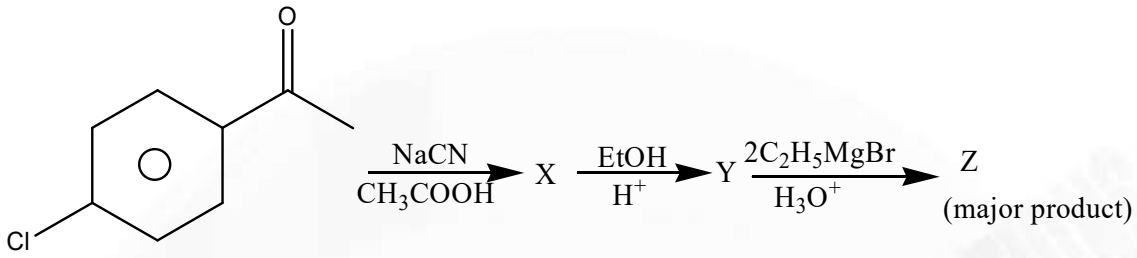
2) 

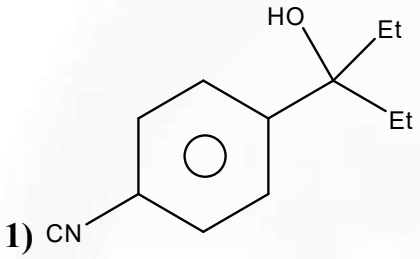
3) 

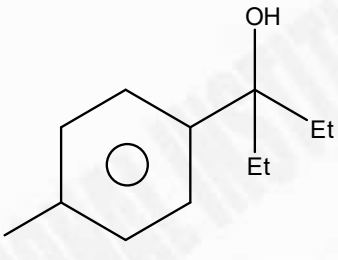
4) 

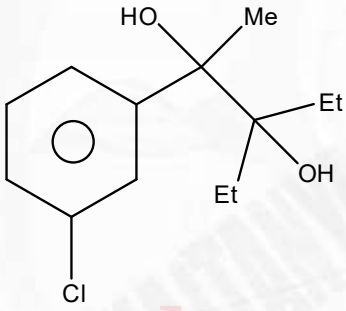
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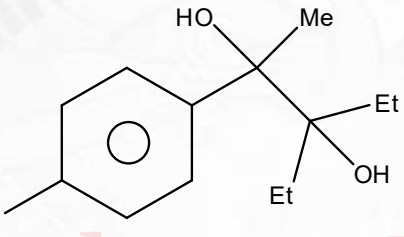
67. Z formed in the following sequence of reactions is:



1) 

2) 

3) 

4) 

68. Given below are two statements

Statement-I : Chlorine has the most negative electron gain enthalpy in its group

Statement-II : Oxygen has the least negative electron gain enthalpy in its group.

In the light of the above statement, choose the most appropriate from the options given below.

1) Statement –I is false but statement –II is true

2) Statement –I and statement –II both are correct

3) statement –I and statement –II both are false

4) Statement –I is true but statement –II is false

69. The P^{ka} of acetic acid and P^{kb} of ammonium hydroxide are 4.76 and 4.75 respectively. The P^{H} of ammonium acetate solution is: (at 298 K)

1) 7 **2)** 4 **3)** 11 **4)** 14

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70. The standard electrode potential for Daniel cell is 1.1V calculate standard Gibbs energy change for the reaction (in kJ/mol)

$Zn(s)+Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)}+Cu(s)$ (1F =96500C)

1) -212 **2)** -106 **3)** -424 **4)** -53

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. Calculate E_{cell} of the cell at 25^0c (nearest integer)

$Mn(s)\left|Mn(OH)_2(s)\right|Mn^{2+}_{(aq)},OH^-\left(10^{-4}M\right)\left|Cu^{2+}(0.675M)\right|Cu(s)$

Given that K_{sp} of $Mn(OH)_2=1.9\times10^{-13}, E^{\Theta}_{Mn^{2+}/Mn}=-1.18V$ and

$$E^{\Theta}_{Cu^{2+}/Cu}=0.34V$$

$2.303\frac{RT}{F}=0.059, \quad \log 2.81=0.45$

72. The total number of paramagnetic species from the following are:

$N_2, O_2, C_2^-, O_2^{2-}, H_2^+, CN^-, He_2^+, O_2^-$

73. A compound (x) with molar mass $108g\,mol^{-1}$ undergoes acetylation to give product with molar mass $192g\,mol^{-1}$ The number of amino groups in the compound (x) per molecule is:

74. For a reaction $X+Y \rightleftharpoons 2Z$, 1.0 mol of X, 1.5 mol of Y and 0.5 mol Z were taken in a 1L vessel and allowed to react. At equilibrium, the concentration of Z was 1.0 $molL^{-1}$. The equilibrium constant of this reaction is $\frac{x}{y}$, the value of $x+y$ is_____.(x and y are integers and x – y = 1).

75. In order to oxidise one mole of each $FeC_2O_4, Fe_2(C_2O_4)_3, FeSO_4$, and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is:

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